
Hodge-Decomposed Preference Optimization: Using the Cycle-Free Component of a Preference Graph as a Training Target

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Abstract

Pairwise preference data can contain cycles ($A \succ B \succ C \succ A$) that no scalar reward function can represent. We propose **Hodge-Decomposed Preference Optimization (HodgePO)**, which applies the combinatorial Hodge decomposition to a preference graph and uses the result in two ways: (1) per-sample weights from how much each response participates in cyclic flow, and (2) a *potential-alignment regularizer* that pulls the model’s reward differences toward the cycle-free (gradient) component of the graph. We evaluate HodgePO at the embedding level on 500 preference pairs from the harmless-base split of Anthropic’s HH-RLHF dataset, over 30 seeds. On ranking accuracy (the fraction of pairs where the preferred response receives the higher reward), Hodge-DPO reaches 0.9999 versus 0.940 for DPO (Cohen’s $d = 6.52$) and Hodge-KTO reaches 0.9964 versus 0.800 for KTO ($d = 16.47$); in both families all 30 paired seed differences are positive. GRPO already reaches the metric ceiling (1.000). We state three limits of this evidence plainly. The accuracy is measured on the training pairs, not on held-out pairs. The cycles in our graph come from similarity-derived edges that we construct, not from observed human judgments. The regularizer’s target is derived from the same labels the metric scores. The result therefore shows that the Hodge gradient potential is an effective in-sample training target. It does not yet show better generalization, and it does not measure robustness to reward hacking.

1. Introduction

Preference optimization methods—Direct Preference Optimization (DPO) (Rafailov et al., 2023), Kahneman-Tversky Optimization (KTO) (Ethayarajh et al., 2024), Group Relative Policy Optimization (GRPO) (Shao et al., 2024), and Odds Ratio Preference Optimization (ORPO) (Hong et al., 2024)—are widely used to align language models with pairwise preference data.

Pairwise preferences are not always globally consistent. The Condorcet paradox (Condorcet, 1785) shows that majority preferences can be cyclic even when every individual preference is transitive, a phenomenon generalized by Arrow’s impossibility theorem (Arrow, 1950). When preference data contains such cycles, no scalar reward function can represent the full preference structure (Jiang et al., 2011). Liu et al. (2025) prove that preferences among LLM responses are representable by a reward model if and only if they contain no Condorcet cycle, and that such cycles arise with probability approaching one under a Luce model. Scalar-reward optimizers must fit such data with some compromise.

Approach. The combinatorial Hodge decomposition (Hodge, 1941; Jiang et al., 2011) splits an edge flow on a graph into a gradient component, which a scalar potential explains exactly, and a cyclic remainder, which no potential explains. We compute this decomposition once on a preference graph and use the gradient potential as a training target for standard preference optimizers.

Contributions.

1. **HodgePO**, a method that augments a pairwise preference optimizer with per-sample cycle-participation weights and a Hodge potential-alignment regularizer (Section 3).
2. Two practical observations: (a) a preference graph with one edge per training pair has no cycles, so cross-pair edges are required for a non-trivial decomposition (Section 3.1); and (b) a batch-level Hodge penalty on

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the predictions of a scalar reward model is identically zero, so the decomposition must be computed on the data, not on the model’s outputs (Appendix C).

3. A 30-seed embedding-level benchmark across seven optimizers (Section 4), reported together with the limits that constrain its interpretation (Section 6).

2. Background

2.1. Preference Optimization

Given a dataset of preference pairs $\mathcal{D} = \{(x, y_w, y_l)\}$ where y_w is preferred over y_l for prompt x , DPO (Rafailov et al., 2023) directly optimizes:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (1)$$

KTO (Ethayarajh et al., 2024) operates on individual samples rather than pairs, using prospect-theoretic loss weighting. GRPO (Shao et al., 2024) uses group-relative advantages with a clipped surrogate objective.

These methods assume preferences are generated by a latent reward function via the Bradley-Terry model (Bradley & Terry, 1952): $P(y_w \succ y_l) = \sigma(r^*(y_w) - r^*(y_l))$. This assumption fails when preferences contain cycles.

2.2. Combinatorial Hodge Theory

Let $G = (V, E, w)$ be a weighted graph with an orientation on each edge, and let K be its clique complex (triangles of G are its 2-cells). An edge flow is a 1-cochain $f \in C^1$. Let $d_0 : C^0 \rightarrow C^1$ be the coboundary, $(d_0\phi)_{ij} = \phi(j) - \phi(i)$, and $d_1 : C^1 \rightarrow C^2$ the curl, $(d_1f)_{ijk} = f_{ij} + f_{jk} + f_{ki}$.

Theorem 1 (Hodge Decomposition (Jiang et al., 2011)). *Every edge flow f decomposes uniquely and orthogonally as*

$$f = \underbrace{d_0\phi}_{\text{gradient}} + \underbrace{d_1^*\psi}_{\text{curl}} + \underbrace{h}_{\text{harmonic}}, \quad (2)$$

with $\phi \in C^0$, $\psi \in C^2$, and $h \in \ker \Delta_1$, where $\Delta_1 = d_0d_0^* + d_1^*d_1$ is the Hodge Laplacian on 1-cochains.

The gradient component $d_0\phi$ is the part of the flow that a global ranking explains. The curl and harmonic components are the cyclic remainder: the curl part consists of inconsistencies local to triangles, and the harmonic part of inconsistencies around longer loops. The first Betti number $\beta_1 = \dim \ker \Delta_1$ is the dimension of the harmonic space.

We use the HodgeRank formulation (Jiang et al., 2011): the gradient potential ϕ^* is the least-squares solution

$$\phi^* = \arg \min_{\phi} \sum_{(i,j) \in E} (f_{ij} - (\phi(j) - \phi(i)))^2. \quad (3)$$

Our implementation adds no 2-cells, so it computes only ϕ^* and the cyclic remainder $r = f - d_0\phi^*$, which contains both the curl and the harmonic parts. Each edge’s flow is the log-odds of its preference probability, clipped to $[-10, 10]$.

3. Hodge-Decomposed Preference Optimization

HodgePO augments a base preference optimizer with two mechanisms derived from a one-time decomposition of a preference graph.

3.1. Preference Graph Construction

A graph with only one direct edge per training pair is a perfect matching when every response appears in exactly one pair. A matching has no cycles, so its decomposition is trivial: the cyclic remainder is zero. All cyclic structure in our graph therefore comes from edges we add between pairs.

Definition 1 (Cross-pair preference graph). *Given N preference pairs with response embeddings $\{(\mathbf{e}_w^{(i)}, \mathbf{e}_l^{(i)})\}_{i=1}^N$, the graph G has:*

- **Direct edges:** (w_i, l_i) for each training pair, with preference probability equal to the pair’s annotation confidence.
- **Similarity edges:** for each preferred response w_i , its $k = 5$ nearest preferred responses w_j (cosine similarity s_{ij}) from other pairs. For each such neighbour we add an edge $w_i \rightarrow w_j$ with probability $\text{clip}(0.5 + 0.2(s_{ij} - 0.5), 0.1, 0.9)$ and an edge $w_i \rightarrow l_j$ with probability $\text{clip}(0.7 + 0.2(1 - s'_{ij}), 0.55, 0.95)$, where s'_{ij} is the cosine similarity between w_i and l_j .

The similarity-edge probabilities are fixed functions of embedding similarity chosen by us. They are not human or model judgments. Because the edge $w_i \rightarrow w_j$ is added from every node toward each of its nearest neighbours, two mutual nearest neighbours receive edges in both directions with probability above 0.5 whenever their similarity exceeds 0.5. The cyclic structure that the decomposition finds is therefore partly a property of this construction. Section 6 discusses the consequences.

3.2. Node-Level Cycle Participation Weights

For each node v , define the cyclic participation score $s(v) = \sum_{e \ni v} |r_e|$, where $r = f - d_0\phi^*$ is the cyclic remainder on edge e . For training pair i :

$$\text{cycle_score}(i) = \frac{s(w_i) + s(l_i)}{\max_j (s(w_j) + s(l_j))} \in [0, 1], \quad (4)$$

and the training weight is

$$\alpha_i = 1 - \lambda_{\text{damp}} \cdot \text{cycle_score}(i), \quad \alpha_i \in [0.4, 1.0], \quad (5)$$

with $\lambda_{\text{damp}} = 0.6$. Pairs whose responses carry more cyclic flow receive lower weight.

3.3. Hodge Potential-Alignment Regularization

The second mechanism is a regularizer that pulls the model’s reward differences toward the gradient potential. For training pair i :

$$\Delta\phi_i = \phi^*(l_i) - \phi^*(w_i). \quad (6)$$

Sign convention. In our flow convention, positive flow on the edge $w_i \rightarrow l_i$ encodes $w_i \succ l_i$, and the gradient component on that edge is $\phi^*(l_i) - \phi^*(w_i)$. For a preference consistent with the global ranking, $\Delta\phi_i > 0$. Using $\phi^*(w_i) - \phi^*(l_i)$ instead gives targets of the wrong sign and trains the model to rank dispreferred responses higher. We note this because it is an easy error to make and it silently inverts the learning signal.

Given the model’s rewards $r_\theta(w_i)$ and $r_\theta(l_i)$, the potential-alignment loss is

$$\mathcal{L}_{\text{Hodge}} = \frac{1}{B} \sum_{i \in \text{batch}} (r_\theta(w_i) - r_\theta(l_i) - \Delta\phi_i)^2, \quad (7)$$

and the total loss for a base method is

$$\mathcal{L}_{\text{HodgePO}} = \mathcal{L}_{\text{base}}(\alpha) + \lambda_{\text{Hodge}} \cdot \mathcal{L}_{\text{Hodge}}, \quad (8)$$

where $\mathcal{L}_{\text{base}}(\alpha)$ is the base loss with per-sample weights α_i .

Note on supervision. The graph contains each training pair’s own direct edge, so $\Delta\phi_i$ carries information from that pair’s label. $\mathcal{L}_{\text{Hodge}}$ is therefore a second supervision signal on the training labels, in addition to $\mathcal{L}_{\text{base}}$, not only a filter on cyclic noise.

Why not a batch-level Hodge penalty? A penalty on the cyclic energy of in-batch reward predictions is identically zero for a scalar reward model, because any set of scalar values defines a consistent ranking (Appendix C). The decomposition must be computed on the data.

4. Experiments

4.1. Setup

Dataset. We use 500 preference pairs from the harmless-base split of Anthropic’s HH-RLHF dataset (Bai et al., 2022). For each pair, the chosen response is the preferred response and the rejected response is the dispreferred response. Every pair has annotation confidence 1.0. Prompts

Algorithm 1 HodgePO Training

Require: Preference pairs $\mathcal{D}$, base optimizer (DPO/KTO/GRPO), λ_{Hodge} , λ_{damp}

- 1: Embed all responses: $\{\mathbf{e}_w^{(i)}, \mathbf{e}_l^{(i)}\}$
- 2: Construct the cross-pair preference graph G (Section 3.1)
- 3: Compute the Hodge decomposition: potential ϕ^* , cyclic remainder r
- 4: Compute per-sample weights α_i (Eq. 5)
- 5: Compute potential differences $\Delta\phi_i$ (Eq. 6)
- 6: **for** each training epoch **do**
- 7: **for** each mini-batch B **do**
- 8: $\mathcal{L} \leftarrow \mathcal{L}_{\text{base}}(\alpha) + \lambda_{\text{Hodge}} \cdot \mathcal{L}_{\text{Hodge}}$
- 9: Update model parameters via $\nabla \mathcal{L}$
- 10: **end for**
- 11: **end for**

and responses are embedded with all-MiniLM-L6-v2 (384 dimensions). The graph is built as in Section 3.1; all pairs fall in one category, so no other cross-pair edges are added. The final edge count was not recorded with the results.

Methods. DPO, KTO, GRPO, ORPO, and the HodgePO variants Hodge-DPO, Hodge-KTO, Hodge-GRPO with $\lambda_{\text{Hodge}} = 0.05$ and $\lambda_{\text{damp}} = 0.6$. All methods use the same model (a 2-layer MLP over the frozen embeddings, hidden dimension 64), learning rate 10^{-3} , batch size 32, and 50 epochs.

Metric. Ranking accuracy: the fraction of the 500 pairs for which the trained model assigns a higher reward to the preferred response than to the dispreferred one. In our code this quantity is named “exploit resistance”, because the dispreferred responses in harmless-base are harmful. *The metric is computed on the same 500 pairs used for training.* It is an in-sample measure of fit. It does not measure generalization to unseen pairs, and it does not measure reward hacking under optimization pressure: there is no environment, no text generation, and no policy optimization against the learned reward.

Statistics. Each method runs 30 seeds (seeds 0–29) on the same data; the Hodge variants share one precomputed graph. We report means, standard deviations, 95% confidence intervals, Welch’s t -tests, and Cohen’s d . Because the Hodge variant and its base method share seeds, we also report the paired seed differences.

4.2. Results

Table 1 and Table 2 give the results.

Table 1. Ranking accuracy on the 500 training pairs (in-sample), 30 seeds per method. “Rel.” is the relative change versus the base method.

Method	Mean	Std	95% CI	Rel.
Hodge-DPO	0.9999	0.0005	± 0.0002	+6.3%
DPO	0.9403	0.0129	± 0.0048	—
Hodge-KTO	0.9964	0.0026	± 0.0010	+24.5%
KTO	0.8004	0.0166	± 0.0062	—
Hodge-GRPO	1.000	0.000	± 0.000	0
GRPO	1.000	0.000	± 0.000	—
ORPO	0.6217	0.0803	± 0.0300	—

Table 2. Hodge variants versus base methods (Welch’s t -test, 30 seeds each).

Comparison	Δ (abs.)	Cohen’s d	p	Paired diffs > 0
Hodge-DPO vs. DPO	+0.060	6.52	2.5×10^{-21}	30/30
Hodge-KTO vs. KTO	+0.196	16.47	5.3×10^{-34}	30/30
Hodge-GRPO vs. GRPO	0.000	—	—	0/30

Hodge-DPO and Hodge-KTO fit the training pairs better than their base methods. Hodge-DPO reaches 0.9999 versus 0.9403 for DPO (absolute +0.060, $d = 6.52$). Hodge-KTO reaches 0.9964 versus 0.8004 for KTO (absolute +0.196, $d = 16.47$). The paired differences are positive for all 30 seeds in both families (DPO family: minimum +0.038, mean +0.060; KTO family: minimum +0.158, mean +0.196).

GRPO is at the ceiling. GRPO and Hodge-GRPO both score exactly 1.000 on every seed. The two cannot be distinguished, and no test is defined for two constant samples. The DPO gain lies in the 6% of headroom below this ceiling.

Ranking of methods. By mean accuracy: $\text{GRPO} = \text{Hodge-GRPO} \geq \text{Hodge-DPO} > \text{Hodge-KTO} > \text{DPO} > \text{KTO} > \text{ORPO}$. GRPO versus Hodge-DPO is not significant ($p = 0.16$, $d = 0.37$). All other adjacent differences have $p < 10^{-7}$.

4.3. Analysis

Cross-pair edges are necessary. Without cross-pair edges, the graph is a matching, the cyclic remainder is zero, $\Delta\phi_i$ equals the flow on pair i ’s own edge, and the cycle weights carry no information.

What we did not test. We did not run a controlled ablation that separates the two mechanisms, and we did not run one that separates the effect of the Hodge projection from

the effect of adding a second label-derived target (Section 3.3). We therefore make no claim about which mechanism produces the gain.

5. Related Work

Preference optimization beyond scalar rewards. DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024), GRPO (Shao et al., 2024), and ORPO (Hong et al., 2024) assume, explicitly or implicitly, a scalar reward. Nash Learning from Human Feedback (Munos et al., 2023) and the minimax-winner approach of Swamy et al. (2024) replace the scalar reward with a preference model and a game-theoretic solution concept, which handles intransitive preferences without decomposing them. The general preference model of Zhang et al. (2025) represents cyclic preferences explicitly. Liu et al. (2025) prove that a reward model exists if and only if the preferences have no Condorcet cycle.

Closest work. Huang et al. (2026) decompose preferences into orthogonal transitive (scalar) and cyclic (vector) components for LLM alignment, building on the game decomposition of Balduzzi et al. (2019). Our work uses a related transitive/cyclic split. It differs in using the combinatorial Hodge decomposition of an explicit preference graph and in using the gradient potential as a regularization target for existing optimizers. The potential/harmonic decomposition of games (Candogan et al., 2011) is the game-theoretic counterpart of the decomposition we use.

Hodge theory in ranking and ML. HodgeRank (Jiang et al., 2011) introduced the combinatorial Hodge decomposition for ranking from pairwise comparisons. Sheaf-theoretic methods appear in graph neural networks (Bodnar et al., 2022); Ayzenberg et al. (2025) survey sheaf theory in deep learning.

Reward overoptimization. Moskovitz et al. (2024) address reward model overoptimization with constrained RLHF over composite reward models, and Coste et al. (2024) use reward model ensembles. These methods address overoptimization of a learned reward during policy optimization, which our benchmark does not include.

6. Discussion and Limitations

The following limits apply to every result in Section 4.

1. In-sample evaluation. The metric is computed on the training pairs. A method can raise it by fitting the training labels more closely without generalizing.

2. The regularizer uses the labels it is scored on. $\Delta\phi_i$ is computed from a graph that contains pair i 's own direct edge. HodgePO therefore trains with an additional label-derived target, and the metric then scores the same labels. Points 1 and 2 together allow a gain that comes from stronger in-sample supervision rather than from removing cyclic noise. A held-out evaluation, and a baseline given an equivalent label-derived target without the Hodge projection, are needed to separate these explanations.

3. The cycles are constructed. Each direct edge connects two responses that appear in no other pair, so the direct edges contain no cycles. All cyclic structure comes from similarity edges whose probabilities are fixed functions of embedding similarity (Section 3.1). The benchmark therefore does not measure how much cyclic structure exists in human preference data, and we make no such claim.

4. Scope. One dataset split, 500 pairs, one embedding model, and a small MLP over frozen embeddings. There is no language-model fine-tuning, no text generation, and no reward-hacking evaluation. The metric saturates (GRPO at 1.000).

5. Method history. An earlier formulation used a batch-level Hodge penalty, which is identically zero for scalar reward models (Appendix C); three earlier 30-seed runs correctly measured it as a no-op. The version reported here changed the regularizer and the graph construction in the same revision, so the reason the earlier version was inert and this one is not has not been isolated beyond the zero-penalty argument.

Next steps. (i) Re-run the benchmark with held-out pairs. (ii) Add a baseline with a label-derived regression target but no Hodge projection. (iii) Measure cyclic structure on data with genuine repeated comparisons (the same response compared with several others by annotators), where cycles are observed rather than constructed. (iv) Evaluate at the language-model level.

7. Conclusion

HodgePO uses the combinatorial Hodge decomposition of a preference graph to produce a cycle-free potential, and uses that potential as a regularization target for standard preference optimizers. In an embedding-level benchmark on 500 HH-RLHF pairs, the regularizer raises in-sample ranking accuracy for DPO and KTO on every one of 30 seeds. The evidence does not yet separate cycle filtering from added supervision, and it does not address generalization or reward hacking; Section 6 lists the experiments that would. We also record two practical points for anyone

applying Hodge methods to preference data: a graph with one edge per pair has no cycles, and a Hodge penalty on the outputs of a scalar reward model is identically zero.

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A. Proof Sketch of the Hodge Decomposition (Theorem 1)

With the weighted inner product $\langle f, g \rangle = \sum_e w_e f_e g_e$ on C^1 and the adjoints d_0^*, d_1^* taken with respect to it, $d_1 d_0 = 0$ implies that $\text{im } d_0$ and $\text{im } d_1^*$ are orthogonal. The finite-dimensional Hodge theorem then gives

$$C^1 = \text{im}(d_0) \oplus \text{im}(d_1^*) \oplus \ker(\Delta_1), \quad \Delta_1 = d_0 d_0^* + d_1^* d_1. \quad (9)$$

The gradient potential solves the normal equations $d_0^* d_0 \phi = d_0^* f$. The solution is unique up to an additive constant on each connected component; we add a ridge term $10^{-4} I$ for numerical stability.

B. Implementation Details

1. Embed prompts and responses with all-MiniLM-L6-v2 (384 dimensions).
2. Add one direct edge per pair, preferred $\rightarrow$ dispreferred, with probability equal to the pair’s confidence.
3. Add similarity edges as in Section 3.1 ($k = 5$ nearest preferred responses by cosine similarity).
4. Convert each edge probability to a log-odds flow (clipped to $[-10, 10]$) and solve the least-squares problem for ϕ^* once, before training.
5. Train a 2-layer MLP ($384 \rightarrow 64 \rightarrow 1$) that maps prompt–response embeddings to a scalar reward. This is not language-model fine-tuning.

Table 3. Hyperparameters.

Parameter	Value
Embedding dim	384
Hidden dim	64
Learning rate	10^{-3}
Epochs	50
Batch size	32
DPO β	0.1
KTO β	0.1
GRPO β	0.04
GRPO group size	8
ORPO λ	0.5
λ_{Hodge}	0.05
λ_{damp}	0.6
k (nearest neighbours)	5
Seeds	30

C. Why a Batch Harmonic Penalty Is Zero

Given batch rewards $\{r_1, \dots, r_{2B}\}$, form the complete graph with edge flow $f_{ij} = r_j - r_i$. The potential $\phi_i = r_i$

reproduces every edge exactly: $(d_0\phi)_{ij} = r_j - r_i = f_{ij}$.
The cyclic remainder is therefore

$$f_{ij} - (d_0\phi)_{ij} = 0 \quad \forall(i, j). \quad (10)$$

Any flow induced by scalar predictions is a pure gradient. The decomposition can detect cycles only in flows that come from outside the model, such as preference data.

D. Full Pairwise Comparison Matrix

Table 4. |Cohen’s d | for all method pairs (30 seeds each), from the same result file as Table 1. Bold: Welch’s $p < 0.05$. GRPO and Hodge-GRPO are both constant at 1.000, so no test is defined between them.

	DPO	H-DPO	GRPO	H-GRPO	KTO	H-KTO	ORPO
DPO	—	6.52	6.54	6.54	9.40	6.02	5.54
H-DPO		—	0.37	0.37	16.96	1.82	6.66
GRPO			—	—	16.98	1.92	6.66
H-GRPO				—	16.98	1.92	6.66
KTO					—	16.47	3.08
H-KTO						—	6.60
ORPO							—

E. Provenance

Every number in Section 4 and Appendix D is taken from the file `shared/results/optimizer_comparison_hodge_v3_30seed.json` in the `shape_of_good_behavior` repository (fields `method_stats`, `pairwise_tests`, and `all_results`; `config.n_samples = 500`). The training pairs are in `shared/results/counterfactual_pairs.json` (500 rows, source `hh_rlhf_direct`, category `harmless-base`). The graph construction is in `shared/src/preference_mapper.py`. The evaluation is the function `evaluate_exploit_resistance` in `shared/src/preference_optimizers.py`, which is called on the training samples.